Lorenzo Palazzetti

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Last update: **October 8, 2026**

Bibliometric Profiles

Scopus	Author Profile	• Publications: **30** • Citations: **333** • H-index: **12**
Google Scholar	Author Profile	• Publications: **31** • Citations: **465** • H-index: **16**
ResearchGate	Author Profile	• Publications: **40** • Citations: **420** • H-index: **14**

Short Biography

I hold a B.Sc. and M.Sc. in Computer Science from the University of Perugia (2018, 2020) and a Ph.D. with distinction from the University of Florence (2024). After a HALY.ID Research Fellowship and a postdoc at Perugia where I built IoT- and computer-vision-based insect-monitoring systems and led research on BVLOS drone autonomy. I am now a Postdoctoral Researcher at Penn State, working on AI- and sensor-driven monitoring for livestock and agriculture. My research sits at the intersection of **algorithm design**, **combinatorial optimization**, **unmanned systems**, **computer vision**, and **IoT**: turning rigorous algorithmic work into field-validated, deployable systems.

Research Interests

Algorithms Design • **Combinatorial Optimization** • **Unmanned Vehicles** • **Computer Vision** • **IoT** • **Ultra-Constrained Systems** • **Edge Computing**.

Education

11/2020–04/2024	**Ph.D., Computer Science.** Thesis: *"Optimizing Drone-Based Applications for Delivery and Smart Agriculture"*. Advisor: Maria Cristina Pinotti, Dept. of Computer Science and Math., University of Florence (Italy). Judgment: ***Excellent with praise***.
12/2018–09/2020	**M.Sc., Computer Science.** Thesis: *"Routing drones being aware of wind conditions: the wind impact on the drone's route"*. Advisor: Maria Cristina Pinotti. Dept. of Computer Science and Math., University of Perugia (Italy). Judgment: ***summa cum laude***.
09/2015–11/2018	**B.Sc., Computer Science.** Thesis: *"Lattice-Based Cryptography"*. Advisor: Marco Baioletti. Dept. of Computer Science and Math., University of Perugia (Italy). Judgment: ***104/110***.
09/2010–07/2015	**Secondary School Diploma in IT.** Technical Pole Franchetti Salviani, Città di Castello (Italy). Judgment: ***100/100***.

Appointments

11/2025–11/2027 **Postdoctoral Researcher.** Pennsylvania State University, PA, USA.
FOCUS: AI- and sensor-driven smart monitoring for poultry and dairy farms.

- **System development:** IoT + computer-vision monitoring systems for data-driven livestock management.
- **IoT integration:** smart-sensor network for continuous environmental and physiological data.
- **Computer vision:** detection and tracking of animal behavior, anomalies, and health indicators.
- **Data & analytics:** pipelines and dashboards for KPIs and early health/welfare detection.

07/2024–11/2025 **Postdoctoral Researcher.** Università degli Studi di Perugia, Perugia, Italy.
FOCUS: computer vision and AI algorithms for autonomous drone navigation in BVLoS mode in GNSS-less environments.

02/2024–NOW **Algorithms and Data Structures Teaching Assistant.** Università degli Studi di Perugia, Perugia, Italy.

11/2023–06/2024 **Haly.ID Project Research Fellow.** Università degli Studi di Perugia, Perugia, Italy.
TITLE: *"IoT data integration with drones: from raw data to activity logbook. Principles of certification"*.

- UAV/IoT protocol definition and integration.
- Data-harvesting and image-elaboration algorithms.
- Trusted logbook design for orchard data with preliminary certification techniques.

10/2019–12/2019 **University Internship.** Advisor: Valentina Poggioni. Project: *Neural Collaborative Filtering: a Framework for Recommendations*. Dept. of Computer Science and Math., University of Perugia.

04/2018–06/2018 **System Manager.** Comune di Umbertide, Perugia, Italy.

03/2014–05/2014 **System Manager.** Comune di Città di Castello, Perugia, Italy.

Qualifications

2026 Italian National Scientific Qualification (ASN). Qualification as Associate Professor in the disciplinary field 01/B1 – Informatics (Jul 2026).

Awards & Honors

Best Paper, 2022 International Conference on Distributed Computing and Networking (ICDCN 2022).

Best Demo, 2022 International Conference on Pervasive Computing and Communications (PerCom 2022).

Travel Grant, 2022 International Conference on Pervasive Computing and Communications (PerCom 2022).

Travel Grant, International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning (DTW 2025).

Best Paper candidate, 2025 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor 2025), Bologna, Italy, 28–30 October 2025.

Grants & Funding

2026 **INdAM–GNCS Research Project.** *"Efficient Routing Algorithms for Drones for Beyond Visual Line of Sight Operability"*.

Research Publications

Journal Articles

(J1) G. Calderone, **L. Palazzetti**, E. Casella, "*An Annotated Aerial Dataset for Training and Benchmarking Computer Vision Models in Turkey Behavior Analysis*", *Springer Nature Scientific Data*, 2026 (accepted).

DOI: **[Scimago Q1 · Scopus Q1 · CiteScore 12.6]**

(J2) **L. Palazzetti**, F. Betti Sorbelli, P. Das, L. Almstedt, D. Niederprüm, L. Wolf, and C. M. Pinotti, "*A comprehensive dataset of Halyomorpha halys for automated detection and monitoring in orchards*", *IEEE Transactions on Networking*, vol. 34, pp. 6664-6677, 2026.

DOI: https://doi.org/10.1109/TON.2026.3722934 **[Scimago Q1 · Scopus Q1 · CiteScore 9.7]**

(J3) **L. Palazzetti**, F. Betti Sorbelli, P. Das, L. Almstedt, D. Niederprüm, L. Wolf, and C. M. Pinotti, "*A comprehensive dataset of Halyomorpha halys for automated detection and monitoring in orchards*", *Springer Nature Scientific Data*, vol. 13, no. 1, Article 1059, Jul. 2026.

DOI: https://doi.org/10.1038/s41597-026-07774-3 **[Scimago Q1 · Scopus Q1 · CiteScore 12.6]**

(J4) **Palazzetti, L.**, Giannetti, D., Verolino, A., Grasso, D.A., Pinotti, C.M., Betti Sorbelli, F. (2025). "*AntPi: A Raspberry Pi based edge–cloud system for real-time ant species detection using YOLO*". *Ecological Informatics*, vol. 91, Art. 103383, Nov. 2025.

DOI: https://doi.org/10.1016/j.ecoinf.2025.103383 **[Scimago Q1 · Scopus Q1 · CiteScore 5.4]**

(J5) Almstedt, L., Baltieri, D., Betti Sorbelli, F., Cattozzi, D., Giannetti, D., Kargar, A., Maistrello, L., Navarra, A., Niederprüm, D., O'Flynn, B., **Palazzetti, L.**, Patelli, N., Piccinini, L., Pinotti, C.M., Wolf, L., Zorbas, D. "*A Comprehensive Pest Monitoring System for Brown Marmorated Stink Bug*". *IEEE Transactions on AgriFood Electronics*, vol. 3, no. 1, pp. 110–120, Mar.–Apr. 2025.

DOI: https://doi.org/10.1109/TAFE.2024.3469538 **[New journal (2023); no Scimago/Scopus quartile or CiteScore yet]**

(J6) **Palazzetti, L.**, Krishnaswamy Rangarajan, A., Dinca, A., Boom, B., Popescu, D., Offermans, P., Pinotti, C.M. "*The hawk eye scan: Halyomorpha halys detection relying on aerial tele photos and neural networks*". *Computers and Electronics in Agriculture*, vol. 226, Art. 109365, 2024.

DOI: https://doi.org/10.1016/j.compag.2024.109365 **[Scimago Q1 · Scopus Q1 · CiteScore 15.1]**

(J7) Almstedt, L., Baltieri, D., Betti Sorbelli, F., Cattozzi, D., Giannetti, D., Kargar, A., Maistrello, L., Navarra, A., Niederprüm, D., O'Flynn, B., **Palazzetti, L.**, Patelli, N., Piccinini, L., Pinotti, C.M., Wolf, L., Zorbas, D. "*Beyond The Naked Eye: Computer Vision for Detecting Brown Marmorated Stink Bug and Its Punctures*". *IEEE Transactions on AgriFood Electronics*, vol. 3, no. 1, pp. 6–17, Mar.–Apr. 2025.

DOI: https://doi.org/10.1109/TAFE.2024.3429537 **[New journal (2023); no Scimago/Scopus quartile or CiteScore yet]**

(J8) Betti Sorbelli, F., Corò, F., Chatterjee, P., Ghobadi, S., **Palazzetti, L.**, Pinotti, C.M. "*A Novel Graph-Based Multi-Layer Framework for Managing Drone BVLoS Operations*". *IEEE Transactions on Network and Service Management*, vol. 21, no. 5, pp. 5091–5105, Oct. 2024.

DOI: https://doi.org/10.1109/TNSM.2024.3401175 **[Scimago Q1 · Scopus Q1 · CiteScore 7.2]**

(J9) Giannetti, D., Patelli, N., **Palazzetti, L.**, Betti Sorbelli, F., Pinotti, C.M., Maistrello, L. "*First use of unmanned aerial vehicles to monitor Halyomorpha halys and recognize it using artificial intelligence*". *Pest Management Science*, vol. 80, Apr. 2024.

DOI: http://dx.doi.org/10.1002/ps.8115 **[Scimago Q1 · Scopus Q1 · CiteScore 7.3]**

(J10) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*Drone-based Bug Detection in Orchards with Nets: A Novel Orienteering Approach*". *ACM Transactions on Sensor Networks*, vol. 20, no. 3, Art. 68, May 2024.

DOI: https://doi.org/10.1145/3653713 **[Scimago Q1 · Scopus Q1 · CiteScore 5.9]**

(J11) Betti Sorbelli, F., **Palazzetti, L.**, Pinotti, C.M. "*YOLO-based Detection of Halyomorpha halys in Orchards Using RGB Cameras and Drones*". *Computers and Electronics in Agriculture*, vol. 213, Art. 108228, 2023.

DOI: https://doi.org/10.1016/j.compag.2023.108228 **[Scimago Q1 · Scopus Q1 · CiteScore 15.1]**

(J12) Betti Sorbelli, F., Navarra, A., **Palazzetti, L.**, Pinotti, C.M., Prencipe, G. "*Wireless IoT Sensors Data Collection Reward Maximization by Leveraging Multiple Energy-and Storage-Constrained UAVs*". *Journal of Computer and System Sciences*, vol. 139, Art. 103475, 2024.

DOI: https://doi.org/10.1016/j.jcss.2023.103475 **[Scimago Q1 · Scopus Q1 · CiteScore 2.6]**

(J13) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*How the Wind Can Be Leveraged for Saving Energy in a Truck-Drone Delivery System*". *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4038–4049, Apr. 2023.

DOI: https://doi.org/10.1109/TITS.2023.3234627 **[Scimago Q1 · Scopus Q1 · CiteScore 13.7]**

(J14) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*On the Scheduling of Conflictual Deliveries in a last-mile delivery scenario with truck-carried drones*". *Pervasive and Mobile Computing*, vol. 87, Art. 101700, 2022.

DOI: https://doi.org/10.1016/j.pmcj.2022.101700 **[Scimago Q1 · Scopus Q1 · CiteScore 7.9]**

Conference and Workshop Proceedings

(C1) F. Betti Sorbelli, **L. Palazzetti**, and C. M. Pinotti, "*Beyond Signal Bars: Benchmarking Cellular RSSI and Quality Mapping from Real-World Traces*", In: *IEEE Global Communications Conference (GLOBECOM 2026)*, Macau S.A.R., China, December 7–11, 2026.

DOI: *Not yet assigned (in press)* **[ICORE B]**

(C2) A. Ceccarelli, J. Colombo, P. Das, F. Gavazzi, S. Ghobadi, M. Kordi, P. Lollini, M. Meo, **L. Palazzetti**, C. M. Pinotti, D. Renga, and G. Vallero, "*Risk- and Connectivity-Aware UAV BVLoS Routing: Design and Experimental Evaluation*", In: *IEEE Global Communications Conference (GLOBECOM 2026)*, Macau S.A.R., China, December 7–11, 2026.

DOI: *Not yet assigned (in press)* [ICORE B]

(C3) Betti Sorbelli, F., Das, P., **Palazzetti, L.**, Pinotti, C.M. "*Can a Classifier Replace the Human? Box-Level Surrogates for Low-Annotation Object Detection*". *22nd DCOSS-IoT*, 2026 (accepted).

DOI: *Not yet assigned (in press)* [ICORE B]

(C4) **Palazzetti, L.**, Molins, M., Casella, E. "*An IoT-Enabled RGB-D Approach for Weight Prediction in Turkey Production*". *22nd DCOSS-IoT*, 2026 (accepted).

DOI: *Not yet assigned (in press)* [ICORE B]

(C5) **Palazzetti, L.**, Chen, K.-L., Casella, E. "*From Drones to Labels: A Semi-Automated Pipeline for Efficient Livestock Segmentation Annotation*". *22nd DCOSS-IoT*, 2026 (accepted).

DOI: *Not yet assigned (in press)* [ICORE B]

(C6) Betti Sorbelli, F., Das, P., **Palazzetti, L.**, Pinotti, C.M. "*EdgeNeXt: A Lightweight Model for UAV-Based Gesture Recognition from Aerial Perspectives*". *IEEE ICC*, 2026 (accepted).

DOI: *Not yet assigned (in press)* [ICORE B]

(C7) Betti Sorbelli, F., **Palazzetti, L.**, Pinotti, C.M. "*Outdoor Accuracy Evaluation of DecaWave's DWM1002 PDoA Kit Measurements*". *IEEE ICC*, 2026 (accepted).

DOI: *Not yet assigned (in press)* [ICORE B]

(C8) Betti Sorbelli, F., Ghobadi, S., **Palazzetti, L.**, Pinotti, C.M. "*Optimizing the Number of Drones for Aerial Power-Line Maintenance*". *21st ALGOWIN*, Warsaw, Poland, 2025.

DOI: https://doi.org/10.1007/978-3-032-09120-8_4 [ICORE C]

(C9) Betti Sorbelli, F., Ghobadi, S., **Palazzetti, L.**, Pinotti, C.M. "*Integrating Ground Communication for Extended Drone Visual Line of Sight*". *21st IEEE WiMob*, Marrakesh, Morocco, pp. 1–6, 2025.

DOI: https://doi.org/10.1109/WiMob66857.2025.11257561 [ICORE C]

(C10) Betti Sorbelli, F., Ghobadi, S., **Palazzetti, L.**, Pinotti, C.M. "*Power Line Maintenance Using Multi-Package Collaborative Drone-Truck System*". *21st IEEE DCOSS-IoT*, 2025. IEEE.

DOI: https://doi.org/10.1109/DCOSS-IoT65416.2025.00061 [ICORE B]

(C11) Betti Sorbelli, F., **Palazzetti, L.**, Pinotti, C.M. "*A Drone-Based Automated Halyomorpha Halys Scouting: A Case Study on Orchard Monitoring*". *IEEE MetroAgriFor*, Pisa, Italy, pp. 380–385, 2023.

DOI: https://doi.org/10.1109/MetroAgriFor58484.2023.10424287 [not CORE-ranked]

(C12) Betti Sorbelli, F., **Palazzetti, L.**, Pinotti, C.M. "*Preliminary Results for Halyomorpha Halys Monitoring Relying on a Custom Dataset*". *IEEE MetroAgriFor*, Pisa, Italy, pp. 363–368, 2023.

DOI: https://doi.org/10.1109/MetroAgriFor58484.2023.10424403 [not CORE-ranked]

(C13) Dinca, A., Popescu, D., Pinotti, C.M., Ichim, L., **Palazzetti, L.**, Angelescu, N. "*Halyomorpha Halys Detection in Orchard from UAV Images Using Convolutional Neural Networks*". *IWANN*, LNCS vol. 14135, Springer, 2023.

DOI: https://doi.org/10.1007/978-3-031-43078-7_26 [ICORE B]

(C14) Almstedt, L., Baltieri, D., Betti Sorbelli, F., Cattozzi, D., Giannetti, D., Kargar, A., Maistrello, L., Navarra, A., Niederprüm, D., O'Flynn, B., **Palazzetti, L.**, Patelli, N., Piccinini, L., Pinotti, C.M., Wolf, L., Zorbas, D. "*Technological Innovations in Agriculture for Scouting Halyomorpha halys in Orchards*". *5th ISIOT Workshop*, Pafos, Cyprus, pp. 702–709, 2023.

DOI: https://doi.org/10.1109/DCOSS-IoT58021.2023.00110 **[co-located with DCOSS-IoT, ICORE B]**

(C15) Betti Sorbelli, F., Chatterjee, P., Corò, F., **Palazzetti, L.**, Pinotti, C.M. "*A Novel Multi-Layer Framework for BVLoS Drone Operation: A Preliminary Study*". *IEEE INFOCOM DroneCom Workshop*, Hoboken, NJ, USA, pp. 1–6, 2023.

DOI: https://doi.org/10.1109/INFOCOMWKSHPS57453.2023.10225806 **[co-located with INFOCOM, ICORE A*]**

(C16) Betti Sorbelli, F., Navarra, A., **Palazzetti, L.**, Pinotti, C.M., Prencipe, G. "*Optimal and Heuristic Algorithms for Data Collection by Using an Energy-and Storage-Constrained Drone*". *18th ALGOSENSORS*, LNCS vol. 13707, Springer, 2022.

DOI: https://doi.org/10.1007/978-3-031-22050-0_2 **[ICORE C (B in earlier rounds)]**

(C17) Betti Sorbelli, F., Corò, F., Das, S.K., Di Bella, E., Maistrello, L., **Palazzetti, L.**, Pinotti, C.M. "*A Drone-based Application for Scouting Halyomorpha halys Bugs in Orchards with Multifunctional Nets*". *IEEE PerCom Workshops*, Pisa, Italy, pp. 127–129, 2022 **(best demo award)**.

DOI: https://doi.org/10.1109/PerComWorkshops53856.2022.9767309 **[co-located with PerCom, ICORE A*]**

(C18) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*Greedy Algorithms for Scheduling Package Delivery with Multiple Drones*". *23rd ICDCN*, New Delhi, India, pp. 31–39, 2022 **(best paper award)**.

DOI: https://doi.org/10.1145/3491003.3491028 **[ICORE B]**

(C19) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*Drone-based optimal and heuristic orienteering algorithms towards bug detection in orchards*". *18th DCOSS*, Marina del Rey, CA, USA, pp. 117–124, 2022.

DOI: https://doi.org/10.1109/DCOSS54816.2022.00032 **[ICORE B]**

(C20) Betti Sorbelli, F., Corò, F., Das, S.K., **Palazzetti, L.**, Pinotti, C.M. "*Cooperative Truck-Drone Scheduling Approach for Last-Mile Deliveries*". *ICTCS (short paper)*, 2021.

DOI: *Not assigned (short paper, no DOI issued)* **[not currently CORE-ranked]**

(C21) **Palazzetti, L.**, Pinotti, C.M., Rigoni, G. "*A run in the wind: Favorable winds make the difference in drone delivery*". *17th DCOSS*, Pafos, Cyprus, pp. 109–116, 2021.

DOI: https://doi.org/10.1109/DCOSS52077.2021.00031 **[ICORE B]**

(C22) **Palazzetti, L.** "*Routing Drones Being Aware of Wind Conditions: a Case Study*". *17th DCOSS*, Pafos, Cyprus, pp. 343–350, 2021.

DOI: https://doi.org/10.1109/DCOSS52077.2021.00062 **[ICORE B]**

Teaching & Mentoring

Teaching

10/2023–01/2024	**Algorithms and Data Structures Tutoring.** Università degli Studi di Perugia, Perugia, Italy. (27 hours)
11/2022–06/2023	**Algorithms and Data Structures Tutoring.** Università degli Studi di Perugia, Perugia, Italy. (57 hours)
03/2022–06/2022	**Algorithms and Data Structures Tutoring.** Università degli Studi di Perugia, Perugia, Italy. (35 hours)
08/2024–12/2024	"**Didactics of algorithms and data structures (including correctness, complexity and classical algorithms)**" course. University of Perugia, Italy. **12 hours / 3 CFU**
03/2026–04/2026	"**Didactics of algorithms and data structures (including correctness, complexity and classical algorithms)**" course. University of Perugia, Italy. **12 hours / 3 CFU**
06/2026	"**Design of Efficient Convolutional Neural Networks for Computer Vision**" Lectures. Pennsylavania State University, PA, USA. **12 hours**

Master Thesis Supervision

(1) (2024) "Cimice Asiatica: Controllo e Monitoraggio con Dashboard Interattive e Algoritmi di Computer Vision su NVIDIA Jetson", Fabio Bocchini, University of Perugia, Italy.

(2) (2025) "Data Structures and Didactic Exercises for Teaching *Algorithms and Data Structures*", Simone Bonacorsi, University of Perugia, Italy.

Bachelor Thesis Supervision

(1) (2025) "Panoramica sugli approcci combinatorici nell'impiego dei droni per le spedizioni e la consegna dei pacchi", Stefano Biagini, University of Perugia, Italy.

Visiting Periods Abroad

07/2023	Technische Universität Braunschweig (TUBS), Braunschweig, Germany.

Invited Talks & Seminars

Talks and Seminars Held

20/02/2025	"*Pest Countermeasures Development Through Digital Twin Orchard Simulations*", Padova, Italy. **seminar**.
05/12/2024	"*Designing UAV-Assisted Systems for Enhanced Data Collection in Wireless Sensor Networks*", Perugia, Italy. **seminar**.
25/11/2024	"*Optimizing Drone-Based Applications for Delivery and Smart Agriculture*", Campus Da Vinci, Umbertide, Italy. **seminar**.
21/03/2024	"*A Gentle Introduction to Computer Vision for Pest Management*", Parma, Italy. **seminar**.
20/03/2024	"*Introduction to Computer Vision*", Perugia, Italy. **class**.
11/07/2023	"*Next-Generation Farming: Drone Technology and Computer Vision to Mitigate Stink Bug Infestations*", Braunschweig, Germany. **seminar**.

10/05/2023	"*Transforming Farming with Drones: Harnessing Data for Precision Agriculture*", Perugia, Italy. **class**.
13/02/2023	"*Halyomorpha Halys Detection using Drone-Captured Images and YOLOv5*", Perugia, Italy. **seminar**.
14/02/2023	"*Exploring the Advancements in Real-Time Object Detection: An Insight into DJI Matrice's Application*", Perugia, Italy. **seminar**.
21/04/2023	"*UNIPG Report: Release of the Latest Drone Application, Machine Learning Advancements, and Preliminary Time Series Analysis*", Perugia, Italy. **seminar**.
16/06/2023	"*Machine Learning Advancements in Haly.ID project*", Perugia, Italy. **seminar**.
14/10/2022	"*A Discussion about Smart Agriculture Technologies for Scouting Halyomorpha halys in Orchards*", Rolla, MO, USA. **seminar**.
05/04/2022	"*Routing Drones Being Aware of Wind Conditions*", Perugia, Italy. **class**.
14/12/2021	"*Drones & their Applications*", Perugia, Italy. **class**.
14/03/2021	"*Transaction & Fork Bitcoin*", Perugia, Italy. **seminar**.
14/03/2021	"*Tor & deanonymization attacks*", Perugia, Italy. **seminar**.

Academic Service

Guest Editor

ACM Journal on Autonomous Transportation Systems (JATS) — Special Issue on *"Navigation for Autonomous Vehicles in Complex and GPS-Denied Environments"*.

General Chair

IEEE Wi-DroIT (2025, 2026).

Co-organizer IEEE MetroAgriFor (2024, 2025).

Technical Program Committee Member

International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT) (2026).

SMARTCOMP Posters/WIP - Demos (2024, 2025, 2026).

International Conference on Pervasive Computing and Communications - Demo Track (PerCom 2025).

International Conference on Pervasive Computing and Communications - PhD Forum (PerCom 2026).

International Conference on Distributed Computing and Networking (ICDCN 2026).

International Conference on Distributed Computing and Networking - Doctoral Symposium (ICDCN 2025).

International Workshop on Unmanned Vehicles and IoT (UAV-IoT 2024, 2025, 2026).

IEEE ASDRoNet (2024).

ALLSENSORS (2024, 2025, 2026).

SENSORCOMM (2024, 2025, 2026).

SENSORDEVICES (2024, 2025, 2026).

International Conference on Accessible Digital Agriculture Technologies (CADAT 2024, 2025, 2026).

International Symposium on Algorithmics of Wireless Networks (ALGOWIN 2025).

International Workshop on Networked Robotics and Communication Systems (NetRobiCS 2025, 2026).

Web Chair

IEEE Wi-DroIT (2023, 2025, 2026).

IEEE ASDRoNet (2024).

Publicity Chair

IEEE ASDRoNet (2024).

IEEE MetroAgriFor (2024).

IEEE Wi-DroIT (2024).

IEEE DCOSS-IoT (2025, 2026).

Reviewer Activity — Journals

Taylor and Francis Geo-Spatial Information Science.

Taylor and Francis Transportmetrica B: Transport Dynamics.

ACM Transactions on Sensor Networks (TOSN).

ACM Journal on Autonomous Transportation Systems (JATS).

F1000Research.

Elsevier Transportation Research Interdisciplinary Perspectives (TRIP).

Elsevier European Journal of Operational Research (EJOR).

Elsevier Expert Systems with Applications (ESWA).

Elsevier Pervasive and Mobile Computing (PMC).

Elsevier Computer and Electronics in Agriculture (CEA).

Elsevier Computers and Operations Research (COR).

Elsevier ARRAY.

Elsevier Smart Agricultural Technology.

Elsevier Ad Hoc Networks.

Elsevier Transportation Research Part C (TR-C).

Elsevier Computer Networks.

Elsevier Reliability Engineering & Systems Safety.

Elsevier Crop Protection.

Elsevier Computer Communications.

IEEE Transactions on Mobile Computing (TMC).

IEEE Transactions on Network Science and Engineering (TNSE).
PAGEPress Journal of Agricultural Engineering (JAE).
Tech Science Press Computers, Materials & Continua (CMC).
Journal of Intelligent and Fuzzy Systems.
MDPI Sensors.
MDPI Aerospace.
MDPI AI.
MDPI Symmetry.
Nature Scientific Reports.
Nature Scientific Data.
Nature Discover Artificial Intelligence.
Nature Discover Applied Sciences.
Nature Discover Plants.

Reviewer Activity — Workshops and Conferences

ACM SIGKDD (2022).
IEEE WoWMoM (2022).
AAMAS (2022).
IEEE MetroAgriFor (2023, 2024, 2025).
IEEE WoWMoM (2024).
IEEE DCOSS (2024, 2026).
IEEE Wi-DroIT (2023, 2024, 2025, 2026).
International Conference on Smart Computing (SMARTCOMP 2024, 2025).
International Workshop on Unmanned Vehicles and IoT (UAV-IoT 2024, 2025).
IEEE ASDRoNet (2024).
CADAT (2024).
ICDCN (2025, 2026).
IEEE NetRobotics (2025, 2026).
International Symposium on Algorithmics of Wireless Networks (ALGOWIN 2025).
Scandinavian Symposium on Algorithm Theory (SWAT 2026).

Conferences and Workshops Attended

The 22nd Annual International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT 2026), Reykjavik, Iceland, June 22-24, 2026 [**presenter**]

7th International Workshop on Wireless Sensors & Drones in IoT Conference Wi-DroIT 2025, Lucca, Italy June 9-11, 2025 [**presenter**]

International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning (DTW), Padova, Italy, February 19-21, 2025 [**presenter**]

Final Workshop on "Studio di scenari multirischio per i disastri naturali nell'area dell'Italia centro-meridionale e della Sicilia", Perugia, Italy, 5 Dec, 2024 [**presenter**]

Demonstration of "Hawk Eye Scan: Halyomorpha Halys Detection Relying on Aerial Tele Photos and Neural Networks", Haly.ID project, Carpi, Italy, 24-25 May, 2024 [**presenter**]

IEEE INFOCOM DroneCom 2023, Drone-Assisted Wireless Communications for 5G and Beyond workshop, New York, USA, 17-20 May, 2023 [**audience**]

Progetto Algoritmi per il controllo distribuito MIcro MAcro robots (MIMA) GCNS INDAM in Pisa, Italy, April 4-6, 2023 [**audience**]

NVIDIA GTC in Virtually, March 21-23, 2023 [**audience**]

2nd Workshop Haly.ID in Perugia, Italy February 13-14, 2023 [**presenter**]

ALGOSENSORS 2022, International Symposium on Algorithms and Experiments for Wireless Sensor Networks, Potsdam, Germany, September 8-9, 2022 [**presenter**]

IEEE DCOSS 2022, International Conference on Distributed Computing in Sensor Systems, Marina Del Rey, LA, California, May 30-June 1, 2022 [**presenter**]

IEEE PerCom 2022, International Conference on Pervasive Computing and Communications, in Pisa, Italy, March 21-25, 2022. [**audience**]

ACM ICDCN 2022, International Conference on Distributed Computing and Networking, in New Delhi, India, January 4-7, 2022 [**audience**]

IEEE DCOSS 2021, International Conference on Distributed Computing in Sensor Systems, in Coral Bay, Pafos, Cyprus, July 14-16, 2021 [**audience**]

ICTCS Italian Conference on Theoretical Computer Science, in Bologna, Italy September 13-15, 2021 [**presenter**].

IEEE Wi-DroIT 2021, International Workshop on Wireless sensors and Drones in Internet of Things, in Coral Bay, Pafos, Cyprus, July 14-16, 2021 [**presenter**].

HALG 2021, Highlights of Algorithms, online, May 31-3 June, 2021 [**audience**].

Professional Affiliations

2024–2025	**Affiliation**, BREADCRUMBS group. Head: Prof. Francesco Betti Sorbelli.
2020–NOW	**Member**, Association for Computing Machinery (ACM).
2022–NOW	**Member**, Institute of Electrical and Electronics Engineers (IEEE).
2020–2024	**Member**, International Project Haly.ID. Head: Prof. Cristina M. Pinotti.
2020–2024	**Affiliation**, Haly.ID Group. Head: Prof. Cristina M. Pinotti.
2020–NOW	**Affiliation**, Research institution (INdAM), GNCS Group. Head: Prof. Alfredo Navarra.

Skills

Computer

OS	Android, Linux, Microsoft Windows.
CODING	Bash, C, C++, Java, JavaScript, LaTeX, MATLAB, Python, PHP, Visual Basic, Visual Basic for Applications, Pascal, OCaml.
FRAMEWORK	Laravel, Bootstrap.
DATABASES	MySQL, PostgreSQL, SQLite.

Web	jQuery, Ajax, JSON, CMS.
Server	Tomcat, Apache.
IDE	PyCharm, NetBeans.
Software	Photoshop, Illustrator, Libreoffice, VirtualBox, Docker, BlueSky.

Languages

Italian	Mother tongue.
English	B2 Level.
French	A2 Level.

Other

Drone license	A1/A3 Open Sub Category.